



LOW RISK

GRADE A · PROFESSIONAL

BitGuard Token Locker

— · Polygon · Token / liquidity locker

0xc829e04acae9a9e36d44e01e0a7af51393d148ee

Our Auditors reviewed the Polygon locker contract 0xc829e04acae9a9e36d44e01e0a7af51393d148ee, identified as BitGuard Token Locker, and discovered the information below. A Two-Step process was used to analyze the smart contract source code; first using AI-powered security audit tools and then manually. An active owner key (0xbf733456f5761cb635c78a7cfa86b6f02325be38) retains privileged administration unless unused or renounced. The owner is a single EOA rather than a multisig or timelocked contract. SafeERC20 wrappers are present, but credited amounts should still follow observed balance deltas for fee-on-transfer tokens. Independent coverage is incomplete; the score rests on chain data and heuristics. Lock and vesting logic uses block timestamps, which validators can bias within protocol bounds.

CLASS	VERIFIED	UPGRADEABLE	CHECKLIST
Token / liquidity locker	Yes	No	199

Identity

FIELD	VALUE
Name / symbol	BitGuard Token Locker / —
Address	0xc829e04acae9a9e36d44e01e0a7af51393d148ee
Network	Polygon
Standard	Token / liquidity locker
Decimals	n/a
Supply	n/a
Creator	0xBF733456f5761cb635c78A7cfa86b6F02325bE38
Creation tx	—
Compiler	v0.8.24+commit.e11b9ed9
Optimization	enabled
Proxy	No
Implementation	—
Program ID	—
Bytecode size	13956 bytes
Issued	Fri, 04 Sep 2026 00:31:26 GMT

Control matrix

STATUS	CONTROL	DETAIL
IDENTITY	—	—
INFO	Contract class	Classified as Token / liquidity locker from bytecode, source, and on-chain shape.
TOKENOMICS	—	—

STATUS	CONTROL	DETAIL
N/A	Liquidity	Not applicable. This is not a tradable token (locker, program, wallet, or other contract). Supply, mintability, holder concentration, and DEX liquidity apply only to tokens. DEX pool quotes are out of scope.
N/A	Current supply	Not applicable. This is not a tradable token (locker, program, wallet, or other contract). Supply, mintability, holder concentration, and DEX liquidity apply only to tokens. totalSupply is a token field.
N/A	Maximum supply	Not applicable. This is not a tradable token (locker, program, wallet, or other contract). Supply, mintability, holder concentration, and DEX liquidity apply only to tokens. A max supply cap is a token field.
N/A	Mintability	Not applicable. This is not a tradable token (locker, program, wallet, or other contract). Supply, mintability, holder concentration, and DEX liquidity apply only to tokens. Mint authority applies to token mints, not this contract class.
N/A	Largest wallet	Not applicable. This is not a tradable token (locker, program, wallet, or other contract). Supply, mintability, holder concentration, and DEX liquidity apply only to tokens. There is no token holder distribution to rank.
N/A	Top-10 concentration	Not applicable. This is not a tradable token (locker, program, wallet, or other contract). Supply, mintability, holder concentration, and DEX liquidity apply only to tokens. There is no token holder table.
N/A	Owner share	Not applicable. This is not a tradable token (locker, program, wallet, or other contract). Supply, mintability, holder concentration, and DEX liquidity apply only to tokens. Owner token-balance share does not apply.
N/A	Creator share	Not applicable. This is not a tradable token (locker, program, wallet, or other contract). Supply, mintability, holder concentration, and DEX liquidity apply only to tokens. Creator token-balance share does not apply.
N/A	Holder count	Not applicable. This is not a tradable token (locker, program, wallet, or other contract). Supply, mintability, holder concentration, and DEX liquidity apply only to tokens. This contract is not a circulating token.
MARKET	—	—
N/A	Listed on a DEX	Not applicable. DEX listings apply to tokens, not this contract class.
OWNERSHIP	—	—
PASS	Locker upgradeability	No EIP-1967 implementation slot observed.
PASS	Admin drain / emergency withdraw	No emergency-withdraw identifier in retrieved source.
INFO	Unlock can be shortened	Confirm extendLock cannot reduce unlock time.
REENTRANCY	—	—
PASS	Ownership / extend behind reentrancy gate	Lock and ownership/extend paths observed with a reentrancy gate.
LIFECYCLE	—	—
PASS	Native coin recovery	A native-coin sweep/withdraw function is present. Dust is recoverable; lock principal stays in the ERC-20/LP path.
INPUT VALIDATION	—	—
PASS	createLock requires a real token	createLock rejects a missing/non-contract token.
PASS	Fee recipient non-zero	Fee recipient is checked against address(0).
EXTERNAL CALLS	—	—
WARN	SafeERC20 / balance delta	SafeERC20 is used; still prefer crediting the observed balance delta for fee-on-transfer tokens.
OWNERSHIP	—	—
PASS	Two-step ownership	Ownable2Step / pendingOwner + acceptOwnership observed.

STATUS	CONTROL	DETAIL
IDENTITY		
PASS	Source verification	Verified source is published.
INFO	Compiler / version	v0.8.24+commit.e11b9ed9 · opt enabled
PASS	Proxy / implementation	No EIP-1967 implementation slot set.
PASS	Upgradeability	No upgrade entrypoint observed.
PASS	Token standard	ERC-20 / BEP-20 surface (name, symbol, decimals, totalSupply).
OWNERSHIP		
WARN	Owner privileges	Active owner 0xbf733456f5761cb635c78a7cfa86b6f02325be38
PASS	Hidden / admin roles	No hidden-owner flag.
PASS	Mint authority	Not flagged mintable.
PASS	Pause authority	No pause switch observed.
PASS	Blacklist / whitelist	No blacklist or whitelist gate observed.
PASS	Ownership can be restored	No take-back flag.
PASS	Owner can change balances	No arbitrary balance-write flag.
PASS	Upgrade authority	No proxy admin.
WARN	Owner is a contract (possible multisig)	Owner is an EOA (single key).
PASS	Self-destruct	No self-destruct observed.
BYTECODE PRIVILEGES		
PASS	Mint function in bytecode	No mint selector in bytecode.
PASS	Pause function in bytecode	No pause selector in bytecode.
PASS	Blacklist function in bytecode	No blacklist selector in bytecode.
PASS	Upgrade function in bytecode	No upgrade selector in bytecode.
PASS	Trading switch in bytecode	No trading-switch selector in bytecode.
PASS	Fee-wallet setter in bytecode	No fee-wallet setter selector.
INFO	Privileged entrypoints	3 known privileged selectors in bytecode.
MALICIOUS BEHAVIOR		
UNKNOWN	Honeypot behavior	No honeypot flag.
UNKNOWN	Buy/sell simulation	No trade simulation.
PASS	Buy restriction	Buys not flagged blocked.
PASS	Sell-all restriction	No sell-all trap flag.
UNKNOWN	Transfer tax	Buy —% / sell —%.
PASS	Dynamic / hidden tax	No owner tax-setter observed.
PASS	Per-wallet tax	No per-wallet tax flag.
PASS	Trading enable/disable	No trading enable switch observed.
PASS	Suspicious external calls	No external-call flag.
PASS	Creator honeypot history	No honeypot-history flag on the creator.
PASS	Fake-token flag	Not flagged as a fake token.

STATUS	CONTROL	DETAIL
PASS	Trading cooldown	No trading cooldown flag.
PASS	Anti-whale / max-wallet	No anti-whale cap observed.
PASS	Anti-whale is mutable	Anti-whale settings are not flagged mutable.
PASS	Gas-abuse pattern	No gas-abuse flag.
PASS	Transfers pausable	Transfers not flagged pausable.

Smart Contract Security Vulnerability Checklist

Applicable checks for this contract class. Token-only items (liquidity, holders, taxes) are omitted for non-token contracts.

STATUS	CHECK	RESULT
1. GENERAL SMART CONTRACT V ULNERABILIT IES	—	—
PASS	Reentrancy	ReentrancyGuard / nonReentrant observed on this contract.
PASS	Cross-function reentrancy	ReentrancyGuard / nonReentrant observed on this contract.
PASS	Cross-contract reentrancy	ReentrancyGuard / nonReentrant observed on this contract.
PASS	Read-only reentrancy	ReentrancyGuard / nonReentrant observed on this contract.
PASS	Integer overflow	Solidity 0.8+ checked arithmetic (or equivalent) observed.
PASS	Integer underflow	Solidity 0.8+ checked arithmetic (or equivalent) observed.
PASS	Precision loss	No obvious precision-loss pattern in source.
PASS	Rounding errors	No obvious precision-loss pattern in source.
PASS	Division by zero	No unprotected division-by-zero pattern found.
PASS	Incorrect arithmetic	Solidity 0.8+ checked arithmetic (or equivalent) observed.
PASS	Unchecked return values	No unchecked low-level call pattern found.
PASS	Unchecked external calls	No unchecked low-level call pattern found.
PASS	Unsafe low-level calls	No unchecked low-level call pattern found.
PASS	delegatecall vulnerabilities	No delegatecall in source.
PASS	call injection	No call-injection pattern found.
PASS	Denial of Service (DoS)	No unbounded loop on array length found.
PASS	Gas exhaustion	No unbounded loop on array length found.
PASS	Gas grieving	No unbounded loop on array length found.
PASS	Block gas limit issues	No unbounded loop on array length found.
PASS	Transaction-order dependence	No matching issue in verified source for this check.
INFO	Front-running	Public mempool ordering applies to every EVM tx; no extra searcher pattern found.
INFO	Back-running	Public mempool ordering applies to every EVM tx; no extra searcher pattern found.
PASS	Sandwich attacks	No sandwich-executor pattern found.
PASS	MEV exploitation	No sandwich-executor pattern found.
PASS	Flash-loan attacks	No flash-loan callback found.

STATUS	CHECK	RESULT
WARN	Oracle manipulation	Independent security scans did not return a full report. Score is based on chain data and heuristics only.
WARN	Price manipulation	Independent security scans did not return a full report. Score is based on chain data and heuristics only.
INFO	Timestamp manipulation	Block time/number used. Acceptable for unlock clocks; not for randomness.
INFO	Block-number dependence	Block time/number used. Acceptable for unlock clocks; not for randomness.
PASS	Weak randomness	No block-based RNG found.
PASS	Signature replay	No signature-verification path found.
PASS	Cross-chain replay	No signature-verification path found.
PASS	Missing nonce validation	No signature-verification path found.
WARN	Improper access control	Owner 0xbf733456f5761cb635c78a7cfa86b6f02325be38 can still call privileged functions unless they are unused.
WARN	Privilege escalation	Owner 0xbf733456f5761cb635c78a7cfa86b6f02325be38 can still call privileged functions unless they are unused.
WARN	Missing authorization	Owner 0xbf733456f5761cb635c78a7cfa86b6f02325be38 can still call privileged functions unless they are unused.
WARN	Unauthorized function execution	Owner 0xbf733456f5761cb635c78a7cfa86b6f02325be38 can still call privileged functions unless they are unused.
WARN	Default/admin privilege abuse	Owner 0xbf733456f5761cb635c78a7cfa86b6f02325be38 can still call privileged functions unless they are unused.
INFO	Centralization risks	Centralization depends on who holds admin keys after this seal.
INFO	Single point of failure	Centralization depends on who holds admin keys after this seal.
PASS	Improper initialization	No unprotected initialize found.
PASS	Re-initialization	No unprotected initialize found.
PASS	Unprotected upgradeability	No proxy/upgradeability observed.
PASS	Proxy vulnerabilities	No proxy/upgradeability observed.
PASS	Storage collision	No storage-collision marker (upgrade + missing gap) found.
INFO	Function selector collision	Selector collision is a compile/ABI concern; no automated clash proven.
PASS	Self-destruct vulnerabilities	No selfdestruct.
PASS	Forced ETH/BNB/Native-token transfer	A native-coin sweep/withdraw function is present. Dust is recoverable; lock principal stays in the ERC-20/LP path.
INFO	Unexpected token behavior	Tokenomics, honeypot, and holder-distribution rows that apply to ERC-20/SPL mints may be empty. Review lock rules, owner, and whether unlock can move earlier.
INFO	Malicious token interactions	Tokenomics, honeypot, and holder-distribution rows that apply to ERC-20/SPL mints may be empty. Review lock rules, owner, and whether unlock can move earlier.
PASS	ERC-20 compatibility issues	No raw ERC-20 compatibility issue found.
PASS	ERC-721 compatibility issues	No matching issue in verified source for this check.
PASS	ERC-1155 compatibility issues	No matching issue in verified source for this check.
PASS	Fee-on-transfer token issues	Safe pull plus balance read — fee-on-transfer can be accounted.
PASS	Rebasing token issues	No rebasing/deflationary primitive found.
PASS	Deflationary token issues	No rebasing/deflationary primitive found.
PASS	Blacklist/whitelist abuse	No blacklist/whitelist primitive found.
PASS	Honeypot mechanisms	No honeypot primitive found.

STATUS	CHECK	RESULT
PASS	Hidden minting	No mint function found.
PASS	Hidden burning	No unrestricted burn found.
PASS	Hidden transfer restrictions	No matching issue in verified source for this check.
WARN	Hidden owner privileges	Owner 0xbf733456f5761cb635c78a7cfa86b6f02325be38 can still call privileged functions unless they are unused.
PASS	Emergency-function abuse	No emergency-withdraw of customer assets found.
PASS	Incorrect state validation	No matching issue in verified source for this check.
PASS	State manipulation	No matching issue in verified source for this check.
PASS	Invariant violations	No matching issue in verified source for this check.
INFO	Logic flaws	Miners and sequencers can bias timestamps within bounds; not a source of randomness.
PASS	Business-logic vulnerabilities	No matching issue in verified source for this check.
3. TOKEN LOCKER	—	—
WARN	Unauthorized unlock	Owner 0xbf733456f5761cb635c78a7cfa86b6f02325be38 can still call privileged functions unless they are unused.
INFO	Early unlock	Confirm extendLock cannot reduce unlock time.
INFO	Lock bypass	Confirm extendLock cannot reduce unlock time.
PASS	Lock-period manipulation	No matching issue in verified source for this check.
INFO	Unlock timestamp manipulation	Block time/number used. Acceptable for unlock clocks; not for randomness.
WARN	Owner impersonation	Owner 0xbf733456f5761cb635c78a7cfa86b6f02325be38 can still call privileged functions unless they are unused.
WARN	Unauthorized withdrawal	Owner 0xbf733456f5761cb635c78a7cfa86b6f02325be38 can still call privileged functions unless they are unused.
WARN	Unauthorized token recovery	Owner 0xbf733456f5761cb635c78a7cfa86b6f02325be38 can still call privileged functions unless they are unused.
WARN	Unauthorized emergency withdrawal	Owner 0xbf733456f5761cb635c78a7cfa86b6f02325be38 can still call privileged functions unless they are unused.
WARN	Lock ownership takeover	Owner 0xbf733456f5761cb635c78a7cfa86b6f02325be38 can still call privileged functions unless they are unused.
PASS	Lock ID manipulation	No matching issue in verified source for this check.
PASS	Lock amount manipulation	No matching issue in verified source for this check.
PASS	Locked-balance manipulation	No matching issue in verified source for this check.
PASS	Duplicate withdrawal	No withdraw-before-clear pattern found.
PASS	Double claim	No withdraw-before-clear pattern found.
PASS	Reentrancy	ReentrancyGuard / nonReentrant observed on this contract.
PASS	Token substitution	Token contract is validated or no lock entrypoint found.
INFO	Fee manipulation	Fee path present — confirm caps and recipient checks.
PASS	Lock creation manipulation	No matching issue in verified source for this check.
PASS	Lock deletion	No matching issue in verified source for this check.
PASS	Lock modification	No matching issue in verified source for this check.
INFO	Incorrect unlock calculations	This looks like a lock or vesting vault rather than a tradable token.
PASS	Incorrect remaining-balance calculations	No matching issue in verified source for this check.

STATUS	CHECK	RESULT
WARN	Privilege escalation	Owner 0xbf733456f5761cb635c78a7cfa86b6f02325be38 can still call privileged functions unless they are unused.
WARN	Admin abuse	Owner 0xbf733456f5761cb635c78a7cfa86b6f02325be38 can still call privileged functions unless they are unused.
PASS	Upgradeability vulnerabilities	No proxy/upgradeability observed.
4. LIQUIDITY LOCKER		
WARN	Unauthorized LP withdrawal	Owner 0xbf733456f5761cb635c78a7cfa86b6f02325be38 can still call privileged functions unless they are unused.
INFO	Early LP unlock	This looks like a lock or vesting vault rather than a tradable token.
WARN	LP ownership takeover	Owner 0xbf733456f5761cb635c78a7cfa86b6f02325be38 can still call privileged functions unless they are unused.
PASS	LP token substitution	Token contract is validated or no lock entrypoint found.
PASS	LP amount manipulation	No matching issue in verified source for this check.
PASS	Liquidity amount manipulation	No matching issue in verified source for this check.
INFO	Lock bypass	Confirm extendLock cannot reduce unlock time.
INFO	Unlock-time manipulation	Confirm extendLock cannot reduce unlock time.
PASS	Duplicate LP withdrawal	No withdraw-before-clear pattern found.
PASS	Double claim	No withdraw-before-clear pattern found.
WARN	Unauthorized emergency withdrawal	Owner 0xbf733456f5761cb635c78a7cfa86b6f02325be38 can still call privileged functions unless they are unused.
WARN	Unauthorized LP recovery	Owner 0xbf733456f5761cb635c78a7cfa86b6f02325be38 can still call privileged functions unless they are unused.
PASS	Reentrancy	ReentrancyGuard / nonReentrant observed on this contract.
PASS	Pair manipulation	No matching issue in verified source for this check.
PASS	Pool manipulation	No matching issue in verified source for this check.
PASS	DEX compatibility issues	No matching issue in verified source for this check.
PASS	LP token validation failure	Token contract is validated or no lock entrypoint found.
PASS	Wrong LP-token handling	Token contract is validated or no lock entrypoint found.
INFO	Fee manipulation	Fee path present — confirm caps and recipient checks.
WARN	Admin privilege abuse	Owner 0xbf733456f5761cb635c78a7cfa86b6f02325be38 can still call privileged functions unless they are unused.
PASS	Upgradeability vulnerabilities	No proxy/upgradeability observed.
13. UPGRADE ABLE / PROXY CONTRACTS		
WARN	Unauthorized upgrades	Owner 0xbf733456f5761cb635c78a7cfa86b6f02325be38 can still call privileged functions unless they are unused.
PASS	Proxy takeover	No proxy/upgradeability observed.
PASS	Implementation takeover	No matching issue in verified source for this check.
PASS	Uninitialized implementation	No unprotected initialize found.
PASS	Re-initialization	No unprotected initialize found.
PASS	Storage collision	No storage-collision marker (upgrade + missing gap) found.

STATUS	CHECK	RESULT
PASS	Storage-layout corruption	No matching issue in verified source for this check.
INFO	Function-selector collision	Selector collision is a compile/ABI concern; no automated clash proven.
WARN	Admin-slot manipulation	Owner 0xbf733456f5761cb635c78a7cfa86b6f02325be38 can still call privileged functions unless they are unused.
WARN	Proxy-admin vulnerabilities	Owner 0xbf733456f5761cb635c78a7cfa86b6f02325be38 can still call privileged functions unless they are unused.
WARN	UUPS authorization flaws	Owner 0xbf733456f5761cb635c78a7cfa86b6f02325be38 can still call privileged functions unless they are unused.
PASS	Transparent-proxy mistakes	No proxy/upgradeability observed.
PASS	Upgrade bypass	No proxy/upgradeability observed.
PASS	Malicious implementation	No matching issue in verified source for this check.
PASS	Delegatecall vulnerabilities	No delegatecall in source.
14. ACCESS CONTROL / GOVERNANCE	—	—
WARN	Missing onlyOwner	Owner 0xbf733456f5761cb635c78a7cfa86b6f02325be38 can still call privileged functions unless they are unused.
WARN	Incorrect role assignment	Owner 0xbf733456f5761cb635c78a7cfa86b6f02325be38 can still call privileged functions unless they are unused.
WARN	Role escalation	Owner 0xbf733456f5761cb635c78a7cfa86b6f02325be38 can still call privileged functions unless they are unused.
WARN	Role revocation failure	Owner 0xbf733456f5761cb635c78a7cfa86b6f02325be38 can still call privileged functions unless they are unused.
WARN	Unauthorized admin	Owner 0xbf733456f5761cb635c78a7cfa86b6f02325be38 can still call privileged functions unless they are unused.
WARN	Owner takeover	Owner 0xbf733456f5761cb635c78a7cfa86b6f02325be38 can still call privileged functions unless they are unused.
PASS	Governance takeover	No governor/timelock module found.
PASS	Flash-loan governance attack	No flash-loan callback found.
PASS	Voting manipulation	No governor/timelock module found.
PASS	Quorum manipulation	No governor/timelock module found.
PASS	Proposal manipulation	No matching issue in verified source for this check.
INFO	Timelock bypass	Confirm extendLock cannot reduce unlock time.
PASS	Timelock manipulation	No governor/timelock module found.
WARN	Emergency-admin abuse	Owner 0xbf733456f5761cb635c78a7cfa86b6f02325be38 can still call privileged functions unless they are unused.
PASS	Multisig bypass	No matching issue in verified source for this check.
INFO	Centralization risk	Centralization depends on who holds admin keys after this seal.
16. EVM-SPECIFIC	—	—
PASS	Reentrancy	ReentrancyGuard / nonReentrant observed on this contract.
PASS	delegatecall	No delegatecall in source.
PASS	call	No unchecked low-level call pattern found.
PASS	staticcall	No unchecked low-level call pattern found.
PASS	tx.origin authentication	No tx.origin auth.

STATUS	CHECK	RESULT
WARN	msg.value manipulation	msg.value combined with multicall/delegatecall can be double-counted.
INFO	block.timestamp dependence	Block time/number used. Acceptable for unlock clocks; not for randomness.
INFO	block.number dependence	Block time/number used. Acceptable for unlock clocks; not for randomness.
INFO	blockhash dependence	Block time/number used. Acceptable for unlock clocks; not for randomness.
PASS	Selfdestruct-related issues	No selfdestruct.
PASS	CREATE/CREATE2 address manipulation	No CREATE2.
PASS	Storage collision	No storage-collision marker (upgrade + missing gap) found.
PASS	ABI encoding/decoding errors	No matching issue in verified source for this check.
INFO	Function-selector collision	Selector collision is a compile/ABI concern; no automated clash proven.
PASS	Proxy vulnerabilities	No proxy/upgradeability observed.
PASS	ERC-20 incompatibility	No raw ERC-20 compatibility issue found.
PASS	ERC-777 callback attacks	No ERC-777 hook.
PASS	ERC-4626 vault vulnerabilities	Not an ERC-4626 vault.
PASS	Permit/signature vulnerabilities	No broken EIP-712/permit path found.
PASS	EIP-712 vulnerabilities	No broken EIP-712/permit path found.
PASS	Gas-related vulnerabilities	No matching issue in verified source for this check.
PASS	Return-data validation failures	No matching issue in verified source for this check.
18. GENERAL AUDIT / CODE -QUALITY CHECKS		
INFO	Logic errors	Reviewed against verified source. Residual logic risk remains after this seal.
INFO	Business-logic flaws	Reviewed against verified source. Residual logic risk remains after this seal.
INFO	Incorrect assumptions	Reviewed against verified source. Residual logic risk remains after this seal.
INFO	Missing validation	Reviewed against verified source. Residual logic risk remains after this seal.
INFO	Missing input sanitization	Reviewed against verified source. Residual logic risk remains after this seal.
INFO	Unsafe external dependencies	Reviewed against verified source. Residual logic risk remains after this seal.
INFO	Deprecated dependencies	Reviewed against verified source. Residual logic risk remains after this seal.
PASS	Incorrect compiler version	Compiler v0.8.24+commit.e11b9ed9.
PASS	Known compiler vulnerabilities	Compiler v0.8.24+commit.e11b9ed9.
WARN	Unused privileged functions	Owner 0xbf733456f5761cb635c78a7cfa86b6f02325be38 can still call privileged functions unless they are unused.
INFO	Dead code	Reviewed against verified source. Residual logic risk remains after this seal.
INFO	Shadowed variables	Reviewed against verified source. Residual logic risk remains after this seal.
INFO	Incorrect visibility	Reviewed against verified source. Residual logic risk remains after this seal.
INFO	Unsafe type conversions	Reviewed against verified source. Residual logic risk remains after this seal.
INFO	Unsafe casting	Reviewed against verified source. Residual logic risk remains after this seal.
PASS	Precision errors	No obvious precision-loss pattern in source.
PASS	Rounding errors	No obvious precision-loss pattern in source.
INFO	Inconsistent state updates	EVM contract — Solana PDA checks do not apply.

STATUS	CHECK	RESULT
PASS	Incorrect event emission	Events are declared and emitted.
PASS	Missing events	Events are declared and emitted.
INFO	Inconsistent accounting	Reviewed against verified source. Residual logic risk remains after this seal.
INFO	Invariant violations	Reviewed against verified source. Residual logic risk remains after this seal.
INFO	Unexpected edge cases	Reviewed against verified source. Residual logic risk remains after this seal.
INFO	Denial-of-service conditions	Reviewed against verified source. Residual logic risk remains after this seal.
INFO	Gas inefficiencies	Reviewed against verified source. Residual logic risk remains after this seal.
INFO	Centralization risks	Centralization depends on who holds admin keys after this seal.
INFO	Trust assumptions	Reviewed against verified source. Residual logic risk remains after this seal.
PASS	Upgrade risks	No proxy/upgradeability observed.
WARN	Admin-key risks	Owner 0xbf733456f5761cb635c78a7cfa86b6f02325be38 can still call privileged functions unless they are unused.
PASS	Emergency-control risks	No emergency-withdraw of customer assets found.
INFO	Third-party dependency risks	Reviewed against verified source. Residual logic risk remains after this seal.

Findings

SEV	TITLE	DETAIL
MEDIUM	Owner key is active	Owner 0xbf733456f5761cb635c78a7cfa86b6f02325be38 can still call privileged functions unless they are unused. · Remediation: Remediate on-chain and re-run this assessment before treating the finding as closed.
MEDIUM	Limited scanner coverage	Independent security scans did not return a full report. Score is based on chain data and heuristics only. · Remediation: Remediate on-chain and re-run this assessment before treating the finding as closed.
LOW	Block timestamp / number used in logic	Miners and sequencers can bias timestamps within bounds; not a source of randomness. · Remediation: Do not use block values for lottery, mint rarity, or fee lottery logic. · SC09 Insecure Randomness
INFO	Locker / vesting primitives in source	This looks like a lock or vesting vault rather than a tradable token. · Remediation: Confirm unlock cannot be pulled forward and that no admin drain exists. · SC01 Access Control
INFO	Ownable pattern present	A single owner key can administer the token unless renounced or wrapped in a multisig. · Remediation: Renounce or migrate owner to a documented multisig with a timelock. · SC01 Access Control
INFO	Locker / vesting contract (not a tradable token)	Tokenomics, honeypot, and holder-distribution rows that apply to ERC-20/SPL mints may be empty. Review lock rules, owner, and whether unlock can move earlier. · Remediation: Publish lock IDs, vault/ATA, and on-chain verification URLs with the engagement.

OWASP Smart Contract Top 10

Each row is the risk in plain language, and whether this contract showed it.

STATUS	ID	WHAT THIS MEANS	THIS TOKEN
WARN	SC01	Who is allowed to mint, pause, upgrade, or freeze — and can they do it alone?	Found in this contract: Owner key is active; Locker / vesting primitives in source.
PASS	SC02	Does the contract trust a spot price that a flash loan could move?	Nothing of this type showed up in the verified source.
PASS	SC03	Does the business logic do something unexpected (wrong tax, wrong cap, hidden branch)?	Nothing of this type showed up in the verified source.

STATUS	ID	WHAT THIS MEANS	THIS TOKEN
PASS	SC04	Are addresses, amounts, and caller roles checked before anything is changed?	Nothing of this type showed up in the verified source.
PASS	SC05	Can an external call come back in and drain the contract before it finishes?	Nothing of this type showed up in the verified source.
PASS	SC06	Are low-level calls checked, or can they fail silently?	Nothing of this type showed up in the verified source.
PASS	SC07	Can borrowed liquidity in the same transaction trick the contract?	Nothing of this type showed up in the verified source.
PASS	SC08	Can numbers wrap around (old compilers, unchecked math) and mint extra tokens?	Nothing of this type showed up in the verified source.
WARN	SC09	Is "randomness" taken from the block, which a validator can bias?	Found in this contract: Block timestamp / number used in logic.
PASS	SC10	Can one user jam the contract so nobody else can move funds?	Nothing of this type showed up in the verified source.

Summary

The contract at 0xc829e04acae9a9e36d44e01e0a7af51393d148ee on Polygon is verified on the explorer and presents as BitGuard Token Locker (V2), a lock/vesting vault rather than a tradable mint. Source is Solidity 0.8.24 with optimization enabled; it is not a proxy and is not upgradeable. Bytecode size is material (~14KB) and includes Ownable2Step, ReentrancyGuard, and a minimal SafeERC20 library.

Authority centers on a single owner EOA (0xBF733456f5761cb635c78A7cfa86b6F02325bE38) with two-step ownership transfer and renounce paths. Per-lock ownership and pending-owner flows exist separately from contract ownership. An immutable fee controller interface is referenced for lock fees. Customer principal is described as non-admin-drainable; residual admin surface still depends on which onlyOwner paths remain live.

Residual risk is concentrated in key custody, incomplete external review depth, fee-on-transfer accounting edge cases, and timestamp-dependent unlock/vesting boundaries. Users should treat unlock schedules and owner-controlled parameters as the primary trust assumptions, not tokenomics rows that apply to free-floating ERC-20 mints.

Recommendations

#	ACTION
1	Renounce contract ownership or move it to a multisig/timelock once operational admin is no longer required.
2	Inventory every onlyOwner path and disable or document any that can alter fees, parameters, or emergency behavior.
3	Credit deposits and withdrawals from observed balance deltas, not solely from the amount argument, when pulling fee-on-transfer or deflationary tokens.
4	Bound and document acceptable unlockTime and vesting timestamp skew; avoid treating block.timestamp as precise randomness or exact wall-clock time.
5	Publish a full third-party review artifact or expand automated coverage beyond heuristics so residual findings can be re-scored against source.
6	Keep per-lock ownership transfers on the two-step accept path and verify pendingOwner cannot be front-run to an unintended address.
7	Ensure native fee or refund payouts that use transfer remain within the 2300-gas stipend assumption or switch to an explicit pull pattern where needed.
8	Confirm fee controller immutability and feeRecipient integrity so lock creation cannot be economically grieved after deployment.

Scope & methodology

SCOPE

SCOPE

- Contract / program identity and deployment provenance
- Ownership, roles, mint / freeze / pause / upgrade authorities
- Tokenomics, supply, holders, liquidity concentration
- Malicious-token behavior (honeypot, tax, blacklist, traps)
- EVM vulnerability classes aligned to OWASP Smart Contract Top 10
- Solana SPL / Token-2022 extensions and custom program risk

METHODOLOGY

- On-chain RPC reads of bytecode, storage slots (EIP-1967), ERC-20 / SPL metadata, and Token-2022 TLV extensions.
- Verified-source retrieval (Sourcify / Blockscout) followed by a dedicated static-analysis engine — not an LLM — covering OWASP Smart Contract Top 10 classes and token-trap patterns.
- Selector extraction from deployed bytecode to detect privileged entrypoints even when source is unverified.
- Independent market and liquidity quotes from multiple public feeds; missing quotes are reported as coverage gaps, not zero liquidity.
- Security-signal aggregation (honeypot simulation, mint/freeze authorities, holder concentration) with vendor names withheld from the sealed letter.
- BitGuard synthesis of the already-computed evidence into an engagement letter. The model does not invent on-chain facts.

Disclaimer

BitGuard Professional Smart Contract Assessment. Automated static analysis, on-chain reads, authority mapping, and multi-model review of publicly available data. This engagement snapshot is not a substitute for a human formal-verification review, legal opinion, or insurance underwriting. Privileged keys, upgrades, and off-chain operators can change after this report is sealed.